Samanyu Arora
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Education

Year	Degree	Institute	GPA
2024–Present	B.S. in Computer Science	University of Wisconsin–Madison	**3.87/4.00**
2023–2024	B.Tech in Mechanical Engineering	IIT Kanpur, India	**Transferred**

Experience

Software Engineering Intern Jan. 2026 – May 2026
Open Source Program Office, University of Wisconsin–Madison

- Developed efficient surrogate models for Matérn covariance computation in Gaussian process workflows, targeting faster repeated kernel evaluations at scale.
- Implemented experimental pipelines to compare analytic transformations, residual interpolation, and asymptotic approximation methods for special-function-heavy numerical routines.
- Benchmarked approximation quality across parameter and distance regimes, identifying stable strategies for reducing computational cost without sacrificing accuracy.

AI Agent Builder Intern Jun. 2025 – Aug. 2025
NeuralSeek

- Designed and deployed intelligent multi-agent architectures integrating LLM orchestration, data pipelines, and structured retrieval workflows within NeuralSeek's proprietary AI framework.
- Conducted system-level evaluations of generative AI deployment pipelines, analyzing model alignment, data-quality metrics, and end-to-end orchestration performance across enterprise-grade infrastructures.
- Implemented and fine-tuned AI agent patterns using function calling, Seek workflows, and data-tuning modules to improve reasoning fidelity and response grounding.

Undergraduate Research Internship, ML & Data Assimilation Aug. 2024 – Jul. 2025
Madison Experimental Mathematics Lab (MXM), UW–Madison Poster Report

- Investigated the integration of Proximal Policy Optimization (PPO) reinforcement learning with Ensemble Kalman Filters (EnKF) for complex dynamical systems including Lorenz 63 and Lorenz 96.
- Trained neural agents to predict accurate system states using EnKF-generated data while reducing computational costs of traditional assimilation.
- Designed reward functions and neural architectures to optimize PPO performance using time-series data with ensemble observations and forecasts.

Research Experience and Publications

Undergraduate Researcher with Professor Grigorios G. Chrysos Mar. 2025 – Present
Department of Electrical and Computer Engineering, UW–Madison

- **Why DDIM Hallucinates More than DDPM: A Theoretical Analysis of Reverse Dynamics**[pdf]
  Muhammad H. Ashiq, **Samanyu Arora**, Abhinav N. Harish, Ishaan Kharbanda, Hung Yun Tseng, Grigorios Chrysos
  International Conference of Machine Learning (ICML 2026) – Accepted
- Theoretically studied hallucination in reverse diffusion dynamics, showing that DDIM trajectories can become trapped between nearby modes while DDPM stochasticity helps samples escape low-density regions.
- Analyzed reverse-time ODE/SDE behavior for Gaussian mixture targets and connected sampler geometry to empirical hallucination-rate differences between deterministic and stochastic diffusion samplers.

Undergraduate Researcher with Professor Christopher J. Geoga May 2025 – Present
Department of Statistics, UW–Madison

- **Fast and Accurate Conditioning for Large-Scale and Online Gaussian Process Prediction Problems**[pdf]
  Samanyu Arora, Christopher J. Geoga
  arXiv:2605.02574 [stat.CO]
- Developed scalable Gaussian process prediction methods that condition on carefully designed linear combinations of data, preserving exact conditional predictive information with far lower computational cost.
- Implemented Julia routines for low-rank cross-covariance compression and fast online prediction over dense grids and large connected test regions.

Research Assistant with Professor Sandeep Silwal Aug. 2024 – May 2025
Department of Computer Sciences, University of Wisconsin–Madison

- Conducted research on optimizing algorithms for F_0 and F_2 estimators, ℓ_p-stable clustering, and Minimum Spanning Tree (MST) problems in differentially private frameworks.
- Developed efficient algorithms capable of processing large-scale datasets while adhering to differential privacy constraints.
- Enhanced the scalability and robustness of clustering methodologies under privacy constraints for machine learning applications.

Selected Projects

Trading System: Web-Based Limit Order Book Simulator | GitHub — Jun. 2025 – Sep. 2025

- **Developed** a real-time limit order book simulator with **live buy/sell matching** using **Django Channels** and **Redis Pub/Sub**.
- **Implemented secure user workflows** including registration, login, password reset, and **role-based access** for placing, modifying, and cancelling orders.
- **Built admin tools** for **bulk user creation via CSV upload** and **CSV export** of order book and trade history, enhancing data management.

StyleGAN Web Application (Synthetic Image Generation) | GitHub — Jan. 2025 – Mar. 2025

- Engineered a web application integrating a **Generative Adversarial Network (GAN)** to produce synthetic images.
- Trained and deployed the GAN model via a **Python Flask backend**, enabling **real-time image generation**.
- Developed a **JavaScript frontend** interfacing with the **Flask backend** for interactive GAN image synthesis.
- Implemented a **RESTful Flask API** with a **/generate endpoint**, providing programmatic access for image generation.

ZK-MONET: Zero Knowledge Proofs for Neural Network Inference — Jan. 2025 – May 2025

- Architected novel framework combining **Polynomial Neural Networks** with **Groth16 SNARKs**, achieving **40% reduction** in proof generation time and reducing complexity from $O(L \cdot m)$ to $O(L)$.
- Developed **Aggregated Polynomial Constraints** compressing thousands of R1CS constraints into 5 checks; designed custom **PN-IOP protocol** with constant-size proofs.
- Implemented **Field-Optimized Quantization** and **sparsity-aware proofs**; achieved **5.43–22.19× constraint reduction**.

Context-Aware RNN Chatbot — Sep. 2024 – Dec. 2024

- Engineered a context-aware chatbot leveraging advanced RNN architectures with **TensorFlow** and **Keras**.
- Integrated state-of-the-art NLP libraries such as **NLTK** and **spaCy** for comprehensive text categorization and sentiment analysis.
- Enhanced model efficiency through memory-augmented deep learning techniques, achieving a 20% increase in training throughput.
- Improved response accuracy through sequence-to-sequence models coupled with attention mechanisms.

GenP: Racing Game with Dynamic Procedural Generation (Unity) | Visual — Sep. 2024 – Dec. 2024

- Built a **Unity (C#)** racing prototype with **spline-guided dynamic mesh generation** for tracks, seedable procedural layouts, and object pooling for stable FPS.
- **Engineered adaptive vehicle physics** (grip, drag, suspension, acceleration curves) that respond to speed, surface type, and track curvature for tighter control.
- Added an **AI Tuning controller** for difficulty scaling—uses player telemetry to place obstacles, tune spawn rates, and adjust pacing

Academic Achievements & Teaching Experience

- Recipient of the **L&S Honors Summer Senior Thesis Research Grant** for proposed senior thesis research on constraint-aware diffusion sampling for reliable scientific generation by College of Letters & Sciences, UW-Madison. Proposal
- Recipient of the **L&S Honors Summer Research Apprenticeship** for proposed research on stochastic trace estimation with oversampled factorizations by College of Letters & Sciences, UW-Madison. Proposal
- Recipient of the **Augusta Schurrer Undergraduate Student Scholarship** for academic excellence and undergraduate achievement in Mathematics by Math Department, UW-Madison.
- Served as **Peer Mentor** for Math 112: College Algebra, supporting students through problem-solving, discussion guidance, and academic mentoring.
- Served as **Peer Mentor** for Comp Sci 240: Intro to Discrete Mathematics, mentoring students in discrete mathematics, proof techniques, recurrence relations, and graph-theoretic reasoning.
- Reviewed research submissions for **CoLorAI 2026**, evaluating work on low-rank representations, tensor methods, parameter-efficient adaptation, probabilistic circuits, and scalable AI systems.
- Reviewed research submissions for **Agentic AI in the Wild: From Hallucinations to Reliable Autonomy 2025**, evaluating work on hallucination, uncertainty, reliability, and verification in large-scale agentic AI systems.
- Named to the **Dean's List** at the University of Wisconsin–Madison in recognition of high academic achievement.
- Secured **All India Rank 2370** in JEE Advanced 2023 among 150k+ candidates.
- Secured **All India Rank 3374** in JEE Main 2023 among 1.2M+ candidates.
- Selected for the **Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship** and the **National Talent Search Examination (NTSE) Scholarship**, two national-level academic distinctions in India recognizing excellence in science and scholastic achievement.

Technical Skills & Coursework

Programming Languages: Python, C/C++, Java, C#, Julia, R, LaTeX
Libraries and OS: NumPy, SciPy, scikit-learn, TensorFlow, PyTorch, Keras, Matplotlib, Pandas, OpenCV, Linux, Windows
Relevant Coursework: Advanced Algorithms*, Reinforcement Learning*, Stochastic Optimization*, Theory of Multi-Agent Systems*, Theoretical Foundations of Machine Learning*, Learning Theory*, Faster Algorithms for Big Data*, Stochastic Processes, Machine Learning
*Graduate courses